

BUSINESS DATA ANALYSIS & VISUALIZATION USING PYTHON

10-Year Financial Performance Analysis of State Bank of India (FY2017–FY2026)

MGN342: Business Analytics | Project 1

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Dataset & Methodology

D Dataset

Source: Screener.in — SBI's audited NSE/BSE filings

Period: FY2017 to FY2026 (10 years)

File: SBI.xlsx — annual Profit & Loss statement

Format: Horizontal (years as columns) — reshaped for analysis

Key Variables Engineered

- $\text{Total Income} = \text{Interest Earned} + \text{Other Income}$
- $\text{Net Interest Income} = \text{Interest Earned} - \text{Interest Expended}$
- $\text{Cost-to-Income \%} = \text{Operating Expenses} \div \text{Total Income}$
- $\text{Net Profit Margin \%} = \text{Net Profit} \div \text{Total Income}$

M Methodology (Pandas & NumPy)

1

Load

Read raw sheet from Google Drive with `pd.read_excel()`

2

Reshape

Transpose so each financial year becomes a row

3

Clean

Drop non-bank rows; convert all columns to numeric

4

Derive

Engineer NII, Cost/Income %, Margin %, CAGR

10.15%

10-Yr Revenue CAGR

45.75%

Avg Cost-to-Income

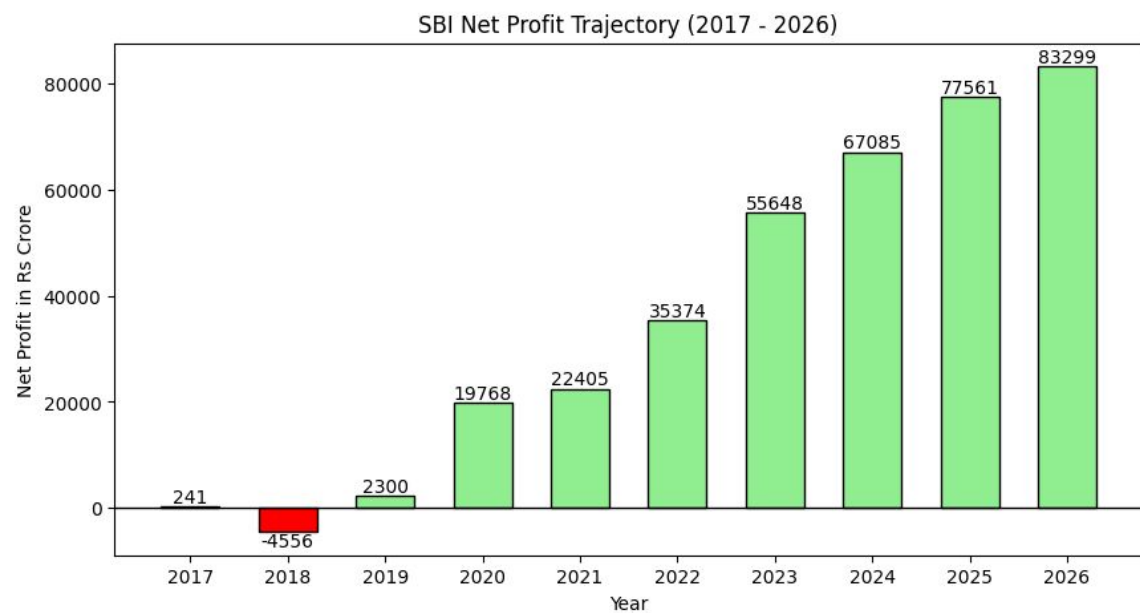
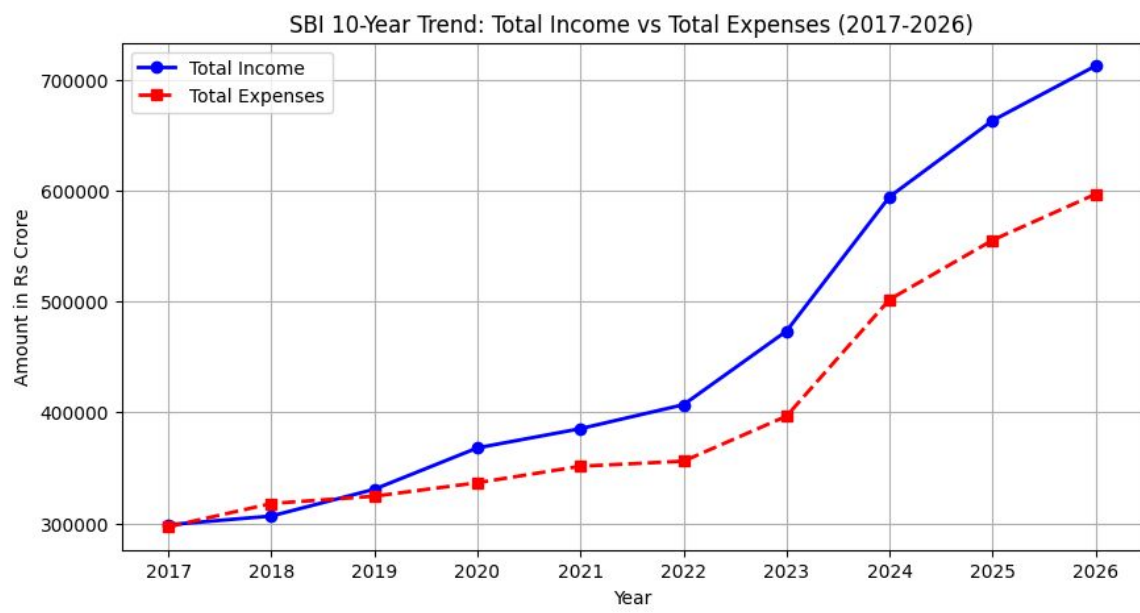
Key Numbers: FY2017 vs FY2026

A decade of steady revenue growth and a dramatic profitability turnaround



The gap between steady revenue growth and explosive profit growth is the core story of this analysis.

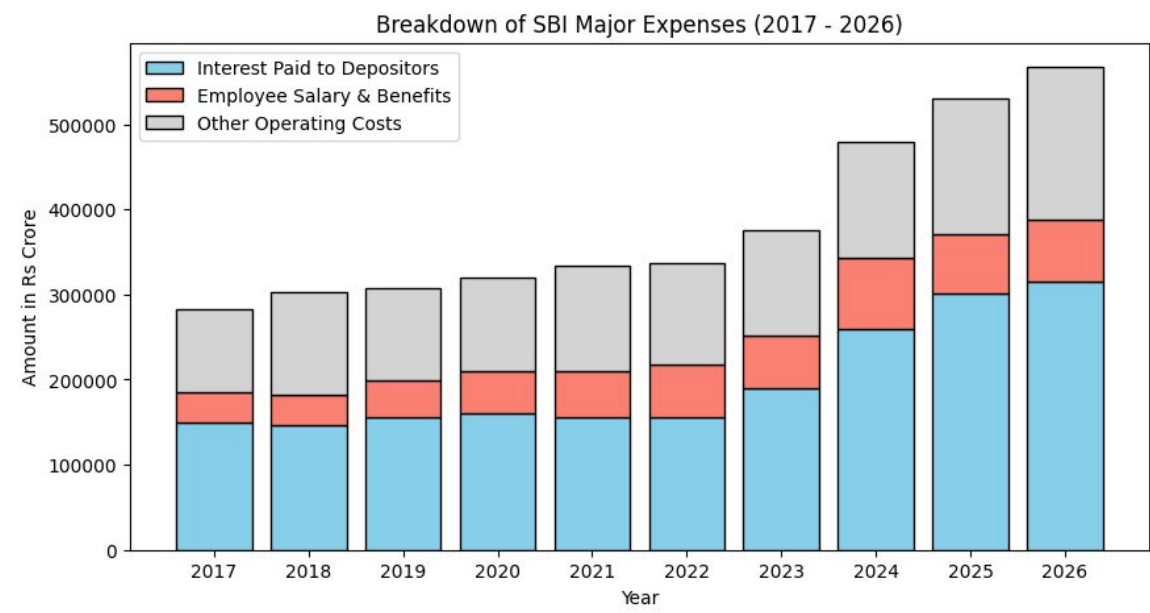
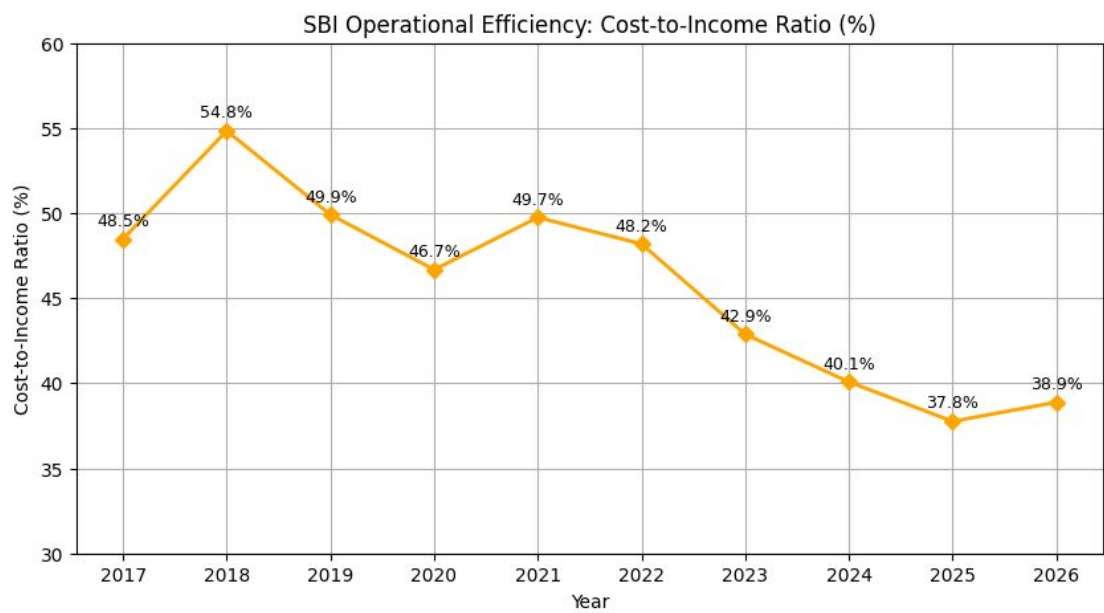
Income, Expenses & Profit Trend



Total Income vs Expenses: Both lines tracked closely until FY2019; the gap widens sharply from FY2021, reaching a ~₹1,14,000 Cr operating surplus by FY2026.

Net Profit Trajectory: A net loss of ₹4,556 Cr in FY2018 (RBI Asset Quality Review) was followed by 8 straight years of profit growth, closing FY2026 at a record ₹83,299 Cr.

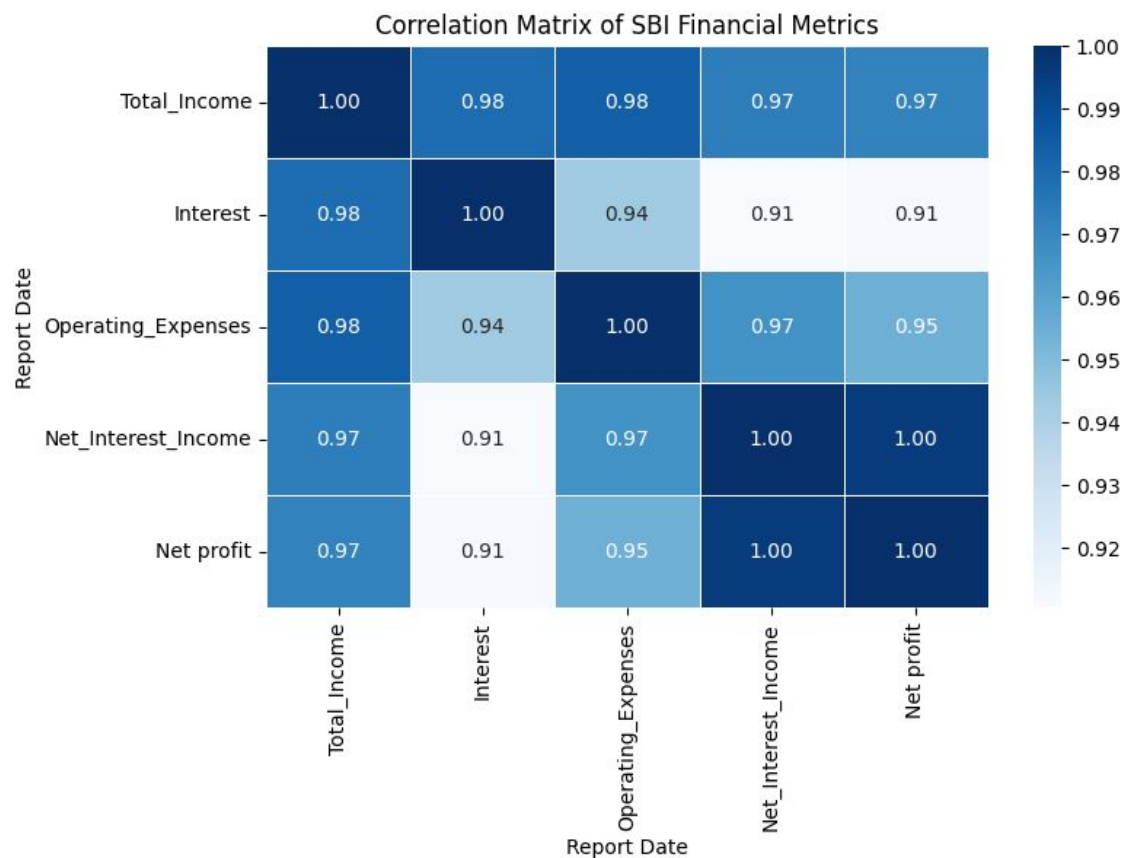
Efficiency & Cost Structure



Cost-to-Income Ratio: Peaked at 54.84% in FY2018, then fell to 37.76% by FY2025 — a ~17-point efficiency gain, likely aided by digital banking (YONO).

Expense Breakdown: Interest paid to depositors is the largest cost every year (>50% of expenses); employee cost grows far more slowly — cost growth is balance-sheet driven, not headcount driven.

Correlation Analysis



What the Data Confirms

$$r = 1.00$$

Net Interest Income ↔ Net Profit

Profit is driven almost entirely by core lending, not fee income.

$$r = -0.94$$

Cost-to-Income Ratio ↔ Net Profit

A strong negative link — years SBI got more efficient are almost exactly the years profit grew fastest.

Findings & Business Insights

FINDINGS

- Revenue grew steadily at a 10.15% CAGR over 10 years
- One loss year (FY2018), then 8 straight years of profit growth
- Cost-to-Income ratio improved by ~17 points from its peak
- Interest expended is the largest cost line every year
- Net Profit is far more volatile than revenue or cost
- Profitability is statistically tied to lending income & efficiency ($r = 1.00 / -0.94$)

BUSINESS INSIGHTS

- The FY2018 loss was a reset, not a decline — it cleared the way for the recovery that followed
- Efficiency gains, not just revenue growth, are driving profit — timing lines up with SBI's YONO digital push
- Interest cost remains the bank's biggest structural exposure to future rate cycles
- Core lending, not fee income, is what actually moves SBI's bottom line

Conclusion

SBI's 10-year story is one of a difficult but necessary reset in FY2018, followed by a sustained, efficiency-led recovery — taking net profit from a loss to over ₹83,000 Crore by FY2026.

Most Useful Statistic

$r = -0.94$ (Cost-to-Income Ratio $\leftrightarrow$ Net Profit)

Thank You

Full notebook, dataset, and report links on the next slide.

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References & Project Links

Click any link below to open the live notebook, dataset, or source

GOOGLE COLAB NOTEBOOK

Full Python code, outputs & visualizations (Pandas, NumPy, Matplotlib, Seaborn)

colab.research.google.com/drive/1KDeVR04ndZDGQbYMsxwhDTXVNfjhjWyP

SBI CLEAN DATASET

Cleaned 10-year Profit & Loss data (FY2017–FY2026) used in the analysis

docs.google.com/spreadsheets/d/1OeyZUSFZlhpqUTaaL9YVeA7I34uzguywOzVQP5bcSDs

ORIGINAL RAW DATASET (SBI.xlsx)

Unprocessed source file, as downloaded from Screener.in

drive.google.com/file/d/1VlqqThZPS0VM0nltqKKmflIvxX6jmvjV

Dataset Source: Screener.in — State Bank of India, Consolidated Financials (audited NSE/BSE filings)